第4章 Cisco IOS介绍

本章目标

通过本章的学习，您应该掌握以下内容:

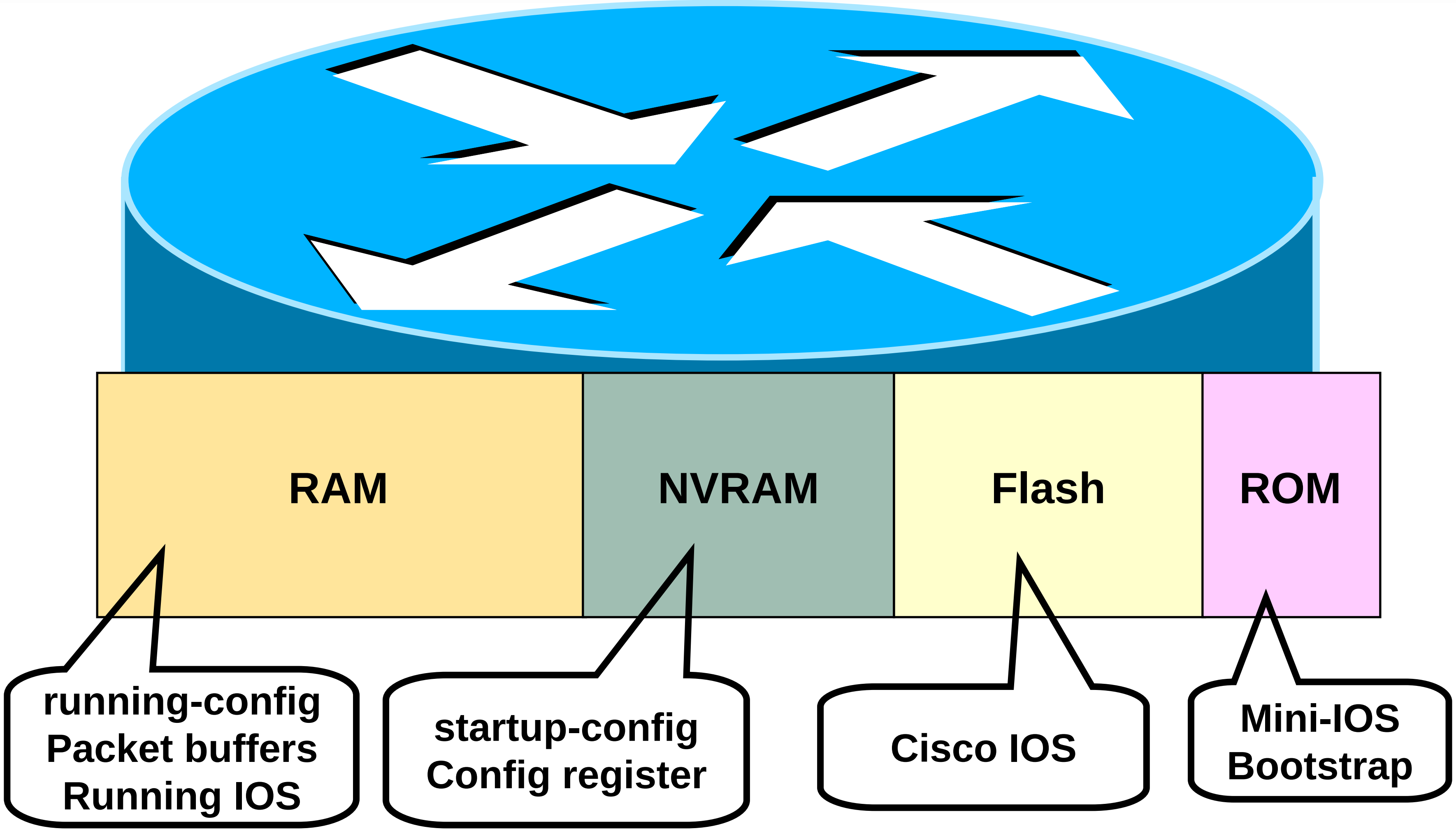
1. 路由器的构造
2. 连接**Cisco**路由器
3. 路由器的初始参数配置
4. 命令行模式
5. 应用帮助和编辑功能
6. **IOS**的常用命令

路由器的构造

路由器由以下几个部分组成：

1. 中央处理单元(**CPU**)
2. 内存
3. 各种接口
4. 操作系统

路由器内存


RAM
NVRAM
Flash
ROM
running-config
Packet buffers
Running IOS
startup-config
Config register
Cisco IOS
Mini-IOS
Bootstrap

路由器的接口

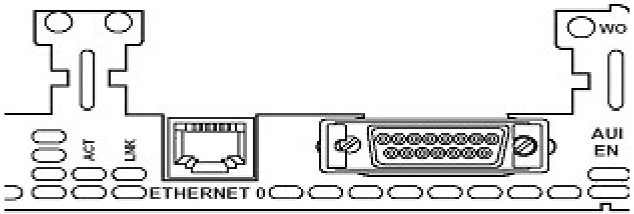
接口类型

1. 广域网接口 **(serial)**
2. 局域网接口 **(ethernet)**
3. 配置接口 **(console)**

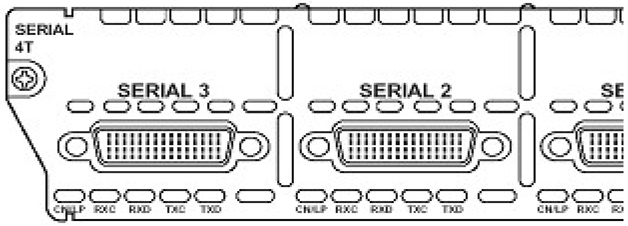
接口命名

1. 全称**+**接口号 **(serial0)**
2. 全称**+**插槽号**+**接口号 **(ethernet1/0)**

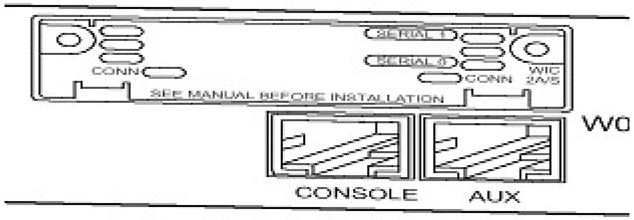
接口



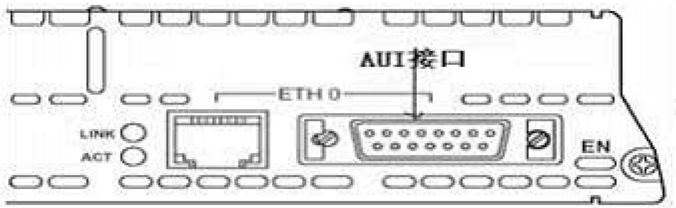
WO
ACT
LNK
ETHERNET 0
AUI
EN



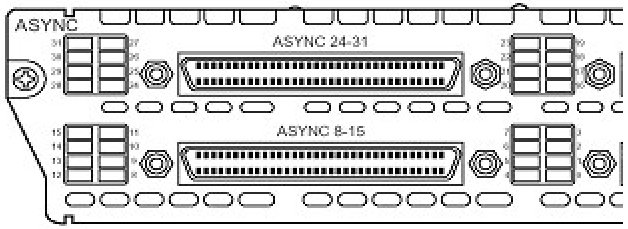
SERIAL
4T
SERIAL 3
SERIAL 2



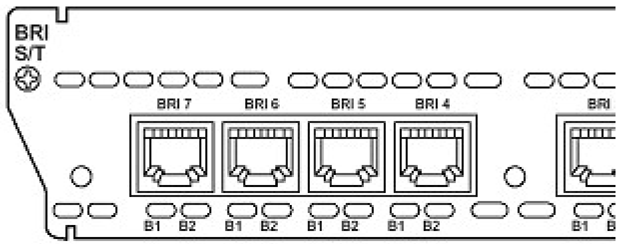
SERIAL 1
SERIAL 0
CONN
CONN
WIC
2A/S
SEE MANUAL BEFORE INSTALLATION
CONSOLE
AUX



AUI接口
ETH 0
LINK
ACT
EN



ASYNC
ASYNC 24-31
ASYNC 8-15



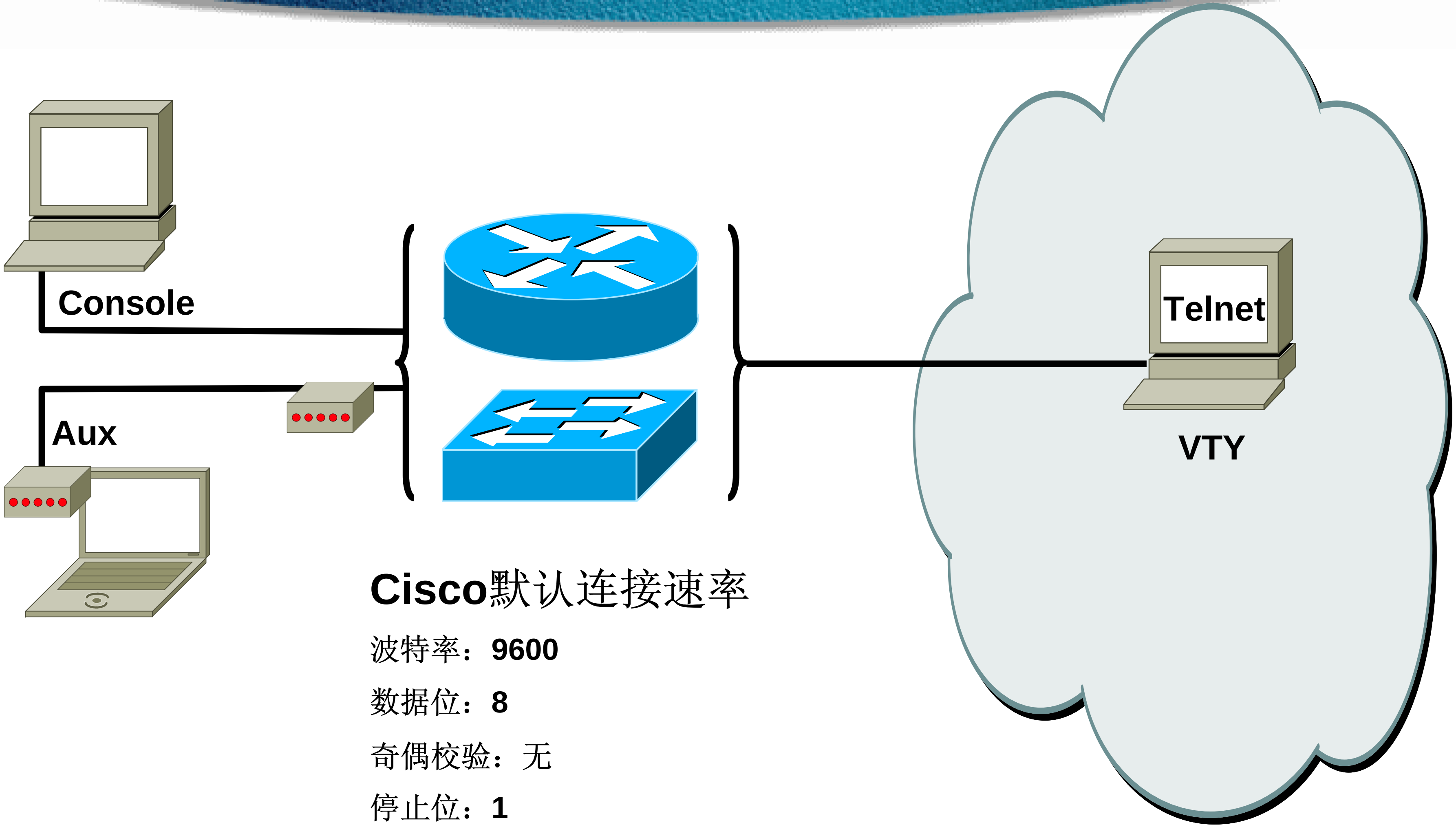
BRI
S/T
BRI 7
BRI 6
BRI 5
BRI 4

路由器操作系统

Internetwork Operating System（**IOS**）是**Cisco**路由器和绝大多数交换机操作系统的内核。

- 加载网络协议和功能。
- 在设备间连接高速流量
- 对未受权的网络使用和访问进行控制增加安全性
- 提供可缩放性，轻松实现网络的增长和冗余要求
- 增强网络资源的可靠性

连接Cisco路由器



Cisco默认连接速率

波特率：**9600**

数据位：**8**

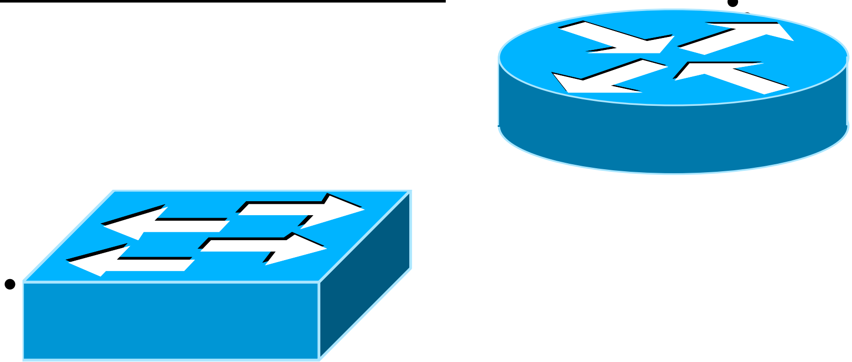
奇偶校验：无

停止位：**1**

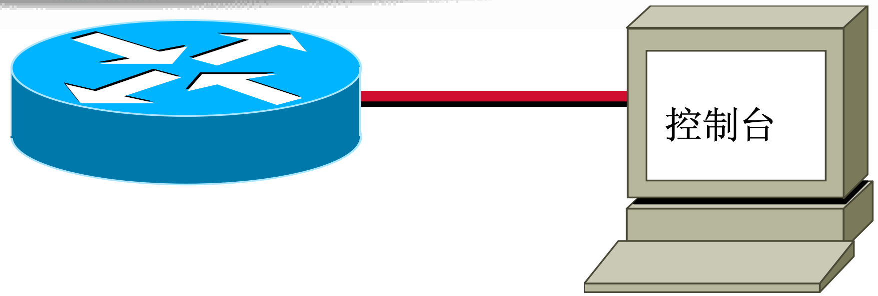
数据流控制：无

Cisco设备的简单启动过程

- 硬件自检（**POST**）
- 定位并加载**Cisco IOS**映象文件（**Flash**）
- 定位并加载配置文件（**NVRAM**）



路由器启动时的输出内容



--- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]:yes

At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[ ]'.

Setup配置方式

```
Router#setup


        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: y

At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[ ]'.


Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system

Would you like to enter basic management setup? [yes/no]: n
```

设置确认与应用

The following configuration command script was created:

hostname Router
enable secret 5 $
enable password
line vty 0 4
password sanjos
no snmp-server
!
no appletalk rout
no decnet routing
ip routing
no clns routing
no ipx routing
no vines routing
no xns routing
no apollo routing
isdn switch-type

interface BRI0
shutdown
no ip address
!
interface Ethernet0
no shutdown
ip address 10.1.1.31 255.255.255.0
no mop enabled
!
interface Serial0
shutdown
no ip address
<text omitted>
end

[0] Go to the IOS command prompt without saving this config.
[1] Return back to the setup without saving this config.
[2] Save this configuration to nvram and exit.

Enter your selection [2]:

命令行模式

Command-line interface（CLI）

- 使用**CLI**拥有更大的配置灵活性；
- 使用**CLI**配置方式功能更强大；
- 使用**CLI**可以抬高你的身价

Cisco IOS的主要命令模式

Router>　用户模式（**user exec mode**）

Router#　特权模式（**privileged exec mode**）

Router(config)#　全局配置模式（**global configuration mode**）

```
Press RETURN to get started.
Router>
Router>enable
Router#
Router#configure terminal
Router(config)#
Router#disable
Router>
Router>logout
```

用户模式提示

特权模式提示

全局配置模式提示

Cisco 接口命令模式

Cisco2611(config)# interface ethernet 0/0
Cisco2611(config-if)#

Cisco2611(config)# interface ethernet 0/0.1
Cisco2611(config-subif)#

Cisco2611(config)#line console 0
Cisco2611(config-line)#

退到特权模式的快捷键：**Ctrl+Z**

路由器命令行帮助机制

上下文关联帮助

提供命令清单和与特定命令相关联的参数.

错误信息提示

指出所输入交换机命令的错误所在，以便于修改或纠正.

之前命令保存区

可以重新调出一个较长或较复杂的命令或其它内容，用来再次运行、查看或修改.

路由器的上下文相关联帮助

```
Router#   clok
Translating "CLOK"
% Unknown command or computer name, or unable to find computer address

Router#   cl?
clear   clock

Router#   clock
% Incomplete command.

Router#   clock ?
set     Set the time and date

Router#   clock set
% Incomplete command.

Router#   <Ctrl-P>clock set ?
hh:mm:ss  Current Time
```

- 符号解析
- 命令提示
- 调出最近用过的内容

增强编辑命令

Ctrl+A	将光标移动到行首
Ctrl+E	将光标移动到行尾
Esc+B	回移一个单词
Ctrl+B	回移一个字符
Ctrl+F	前移一个字符
Esc+F	前移一个单词
Ctrl+D	删除一个字符
Backspace	删除一个字符
Ctrl+R	重新显示一行
Ctrl+U	删除一行
Ctrl+W	删除一个单词
Ctrl+Z	结束配置模式，返回**EXEC**模式
Tab	自动补全命令

回览之前用过的命令

Ctrl-P or Up arrow	调出最近(前一)使用过的命令
Ctrl-N or Down arrow	调出更近使用过的命令
Router> show history	显示命令保存区内容（默认**10**个）
Router> show terminal	显示终端当前历史命令缓存大小
Router> terminal history size *lines*	设置历史命令缓存大小（最大**256**）

系统默认保存10条命令，最大可以设置到256

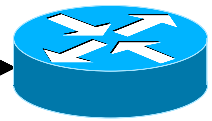
show version 命令
收集路由器基本信息

```
Router5# show version
Cisco Internetwork Operating System Software
IOS (tm) C2600 Software (C2600-I-M), Version 12.2(8)T10,  RELEASE SOFTWARE (fc1)
TAC Support: http://www.cisco.com/tac
Copyright (c) 1986-2003 by cisco Systems, Inc.
Compiled Fri 30-May-03 02:45 by kellythw
Image text-base: 0x80008074, data-base: 0x80A2C2B0

ROM: System Bootstrap, Version 12.2(7r) [cmong 7r], RELEASE SOFTWARE (fc1)

Router2621 uptime is 6 weeks, 1 day, 5 hours, 38 minutes
System returned to ROM by power-on
System image file is "flash:c2600-i-mz.122-8.T10.bin"

cisco 2621XM (MPC860P) processor (revision 0x100) with 93184K/5120K bytes of mem
ory.
Processor board ID JAE07450GQ1 (906407731)
M860 processor: part number 5, mask 2
Bridging software.
X.25 software, Version 3.0.0.
2 FastEthernet/IEEE 802.3 interface(s)
32K bytes of non-volatile configuration memory.
32768K bytes of processor board System flash (Read/Write)

Configuration register is 0x2102
```

路由器密码配置（特权密码）

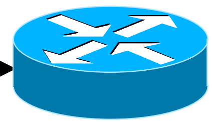
特权模式明文密码

Router(config)# enable password cisco2004



特权模式加密密码

Router(config)# enable secret cisco

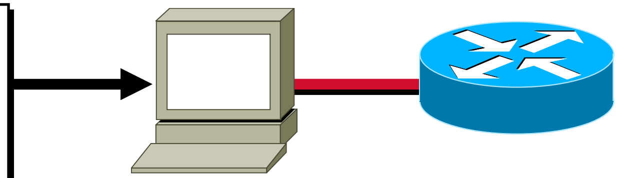


如果两个密码相同，系统将会提示更改。而且一旦设置了加密密码，明文密码就会失效

路由器密码配置（接口密码）

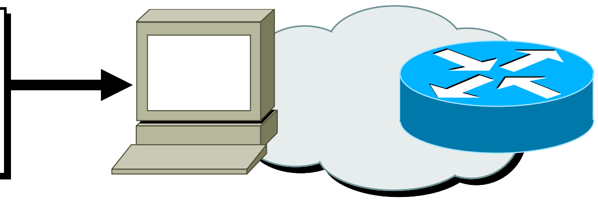
控制台/**Auxiliary**密码

```
Router(config)# line console 0 / line aux 0
Router(config-line)# login
Router(config-line)# password cisco
Router(config-line)# exec-timeout 0 0
Router(config-line)# logging synchronous
```



虚拟终端密码

```
Router(config)# line vty 0 4
Router(config-line)# login
Router(config-line)# password cisco
```



密码是大小写敏感的！

如果要用**telnet**远程登录来管理路由器，则必须设置**vty**线路的密码，否则不能用！ 或使用“**no login**”

路由器密码配置（加密明文密码）

执行密码加密操作

```
Router(config)# service password-encryption
Router(config)# no service password-encryption
```

1. 在命令**service password-encryption**之后设置的密码使用密文存放，在命令**no service password-encryption** 之后设置的密码使用明文存放。
2. 不过，当你输入了**no service password-encryption** 命令之后，已经加密的密码不会重新变成明文，而是你再改的密码会以明文存放。
3. 可用**show run** 命令来查看是否启用口令加密。

Banner设置

MOTD Banner的设置中应该注意**“delimiting character”**的使用。
格式为：

Cisco2611(config)# banner motd @
Enter TEXT message. End with the character '@'.
Welcome to NetLab! HaHa!
@

Cisco2611(config)#

还有三种**Banner**：
Exec banner
Incoming banner
Login banner

路由器接口有关命令

启动接口

Cisco2611(config-if)# no shutdown

关闭接口

Cisco2611(config-if)# shutdown

为接口分配**IP**地址

Cisco2611(config-if)# ip address 172.16.10.2 255.255.255.0

Cisco2611(config-if)# ip address 172.16.10.4 255.255.255.0 secondary

Cisco2611(config)# int serial 0/0

Cisco2611(config-if)# clock rate 64000 (DCE：show controllers serial 0/0)

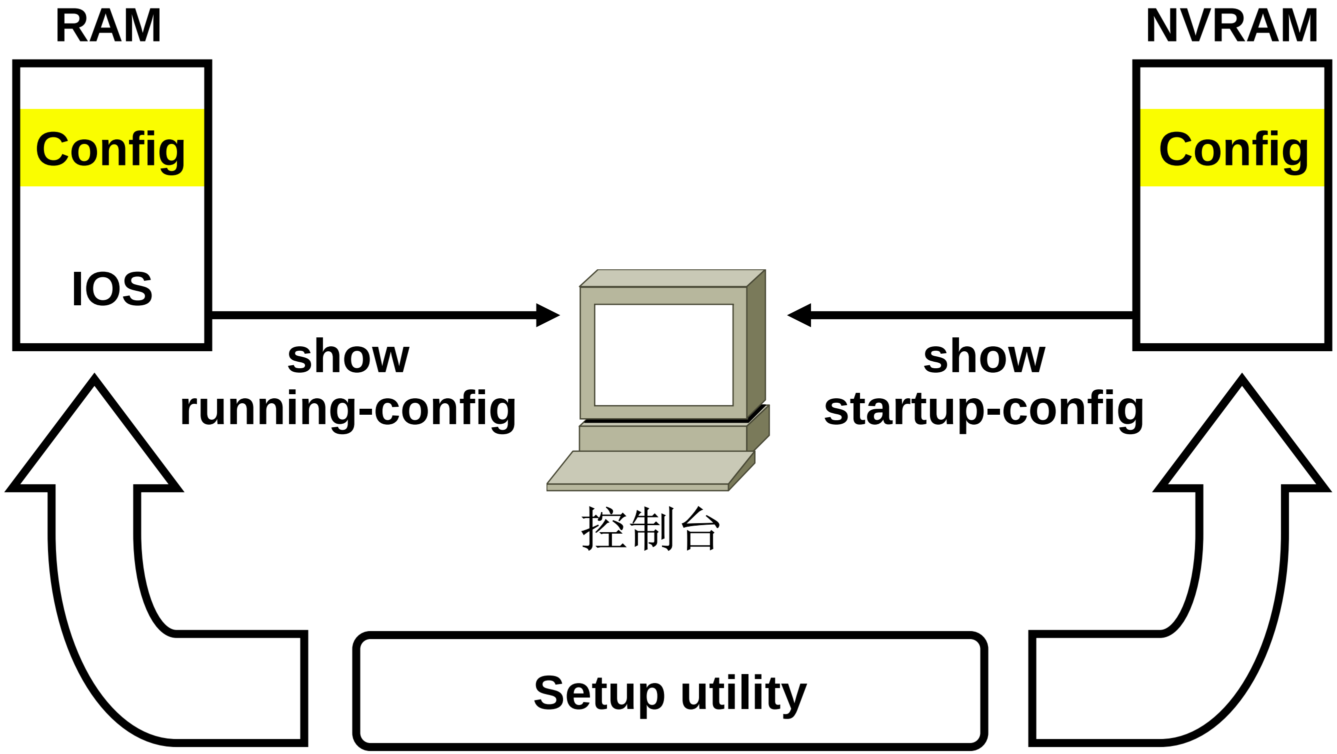
Cisco2611(config-if)# bandwidth 64 (单位是KB)

Cisco2611(config)# hostname Derek

Derek(config)#

Derek(config-if)# description lan interface

查看配置情况



将配置参数保存到NVRAM中

保存当前配置

```
Router2611# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router2611#
```

将当前运行的配置复制到**NVRAM**中，也可用**erase startup-config**进行删除。**Reload**命令用于重启。

路由器的 *show interfaces* 命令

```
Router#show interfaces
Ethernet0 is up, line protocol is up
  Hardware is Lance, address is 00e0.1e5d.ae2f (bia 00e0.1e5d.ae2f)
  Internet address is 10.1.1.11/24
  MTU 1500 bytes, BW 10000 Kbit, DLY 1000 usec, rely 255/255, load 1/255
  Encapsulation ARPA, loopback not set, keepalive set (10 sec)
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:07, output 00:00:08, output hang never
  Last clearing of "show interface" counters never
  Queueing strategy: fifo
  Output queue 0/40, 0 drops; input queue 0/75, 0 drops
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     81833 packets input, 27556491 bytes, 0 no buffer
     Received 42308 broadcasts, 0 runts, 0 giants, 0 throttles
     1 input errors, 0 CRC, 0 frame, 0 overrun, 1 ignored, 0 abort
     0 input packets with dribble condition detected
     55794 packets output, 3929696 bytes, 0 underruns
     0 output errors, 0 collisions, 1 interface resets
     0 babbles, 0 late collision, 4 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
```

解读端口状态

```
Router#show interfaces serial 1

Serial1 is up, line protocol is up

   Hardware is HD64570
   Description: 64Kb Line to San Jose
  ::  ::  ::  ::  ::  ::  ::  ::  ::
```

Operational................. Serial1 is up, line protocol is up
Connection problem... Serial1 is up, line protocol is down
Interface problem........ Serial1 is down, line protocol is down
Disabled .... Serial1 is administratively down, line protocol is down

Clearing *show interface* Counters

```
Router# clear counters
```

```
Router# show interface serial 1

Serial1 is up, line protocol is up

  Hardware is cxBus Serial
  Description: 56Kb Line San Jose - MP
  Internet address is 150.136.190.203, subnet mask is 255.255.255.0
  MTU 1500 bytes, BW 56 Kbit, DLY 20000 usec, rely 255/255, load 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input 0:00:07, output 0:00:00, output hang never
  Last clearing of "show interface" counters 2w4d
  Output queue 0/40, 0 drops; input queue 0/75, 0 drops
  Five minute input rate 0 bits/sec, 0 packets/sec
  Five minute output rate 0 bits/sec, 0 packets/sec
     16263 packets input, 1347238 bytes, 0 no buffer
     Received 13983 broadcasts, 0 runts, 0 giants
     2 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 2 abort
     0 input packets with dribble condition detected
     22146 packets output, 2383680 bytes, 0 underruns
     0 output errors, 0 collisions, 2 interface resets, 0 restarts
     1 carrier transitions
```

Cisco 2621 fa0/1

```
Router2621#sh int fa0/1
FastEthernet0/1 is up, line protocol is up
  Hardware is AmdFE, address is 000e.38ee.3401 (bia 000e.38ee.3401)
  Internet address is 202.114.78.52/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 3/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, 100BaseTX/FX
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:00, output 00:00:00, output hang never
  Last clearing of "show interface" counters never
  Input queue: 1/75/17263/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
```

```
5 minute input rate 1406000 bits/sec, 187 packets/sec
  5 minute output rate 307000 bits/sec, 142 packets/sec
     1797914249 packets input, 4163406475 bytes
     Received 17679119 broadcasts, 0 runts, 0 giants, 0 throttles
     158648 input errors, 0 CRC, 0 frame, 0 overrun, 107328 ignored
     0 watchdog
     0 input packets with dribble condition detected
     2071304159 packets output, 264437907 bytes, 0 underruns
     0 output errors, 0 collisions, 1 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
Router2621#
```

串口上用 *show controller* 命令

```
Router#show controller serial 0
HD unit 0, idb = 0x121C04, driver structure at 0x127078
buffer size 1524  HD unit 0, V.35 DTE cable
                    .
                    .
                    .
```

显示串口线的线缆类型

本章总结

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6. IOS的常用命令

END